

Nirvan Rabbani

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EDUCATION

University of Waterloo

Bachelor of Software Engineering, Honours, Co-operative Program

GPA 3.92/4.00

- Relevant Coursework: Object-Oriented Programming, SDLC Methodologies, Embedded Systems, Data Structures

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript/TypeScript, Go, Rust, SQL, HTML/CSS
Frameworks: React/React Native, Redux, Node.js/Express, Next.js, Django, FastAPI, PyTorch, Firebase, LangGraph
Databases: Oracle DB, PostgreSQL, Supabase, SQLite, MySQL, MongoDB, Redis, pgvector
Tools: Git, Linux, Jira, Docker, Linear, GitHub Actions, Kubernetes, Google Cloud, Azure, GTest, Claude Code

RESEARCH

SwiftQL – Independent Query Engine Research 🌀 | C++17, Python, CMake, GoogleTest, SQLite, TPC-H

- Built a **SQL** engine from scratch in **C++**; raw query text into vectorized plans over a custom columnar storage
- Achieved **22/22 TPC-H** benchmark queries at SF1, beating **SQLite 3.0×** and 1-thread **PostgreSQL 2.0×**
- Cut memory **6×** with dictionary/RLE encodings and skipped **93%** of chunks on sorted data via zone-map pruning
- Traced 3 slowest operators to row-at-a-time loops inside vectorized batches; rewriting them cut latency by **26%**
- Made queries **2.22×** faster with a cost-based optimizer that orders joins by **DP** and picks a **SIMD** or hash join

EXPERIENCE

AltaML

May 2026 – Aug. 2026

Associate Software Developer, Backend / AI Infrastructure

Toronto, ON

- Shipped agentic spam moderation pipeline, automating **1.8K/2K** daily items, cutting review time **10h → 1h**
- Built a **LangGraph** ReAct agent with hybrid **RAG** achieving **94.3%** classification accuracy on leak-free evals
- Engineered **BigQuery ETL** powering **30+ MCP** tools serving ~20 live calls/item across **100GB+** of forum data
- Reconciled queue data drift with site-scoped keys & **Postgres** advisory locks, preventing duplicate processing
- Built **GitHub Actions** CI/CD to run E2E tests on every PR via **Playwright** saving **200+** hours of manual QA

Syllavise Inc.

Jan. 2026 – Aug. 2026

Founding Engineer

Remote

- Led 3 engineers on **Python/FastAPI + React Native** academic planning SaaS shipped to web, iOS & Android
- Built syllabus ETL: **LlamaParse** → **Claude** → typed schema, **>90%** field F1 on 300 labeled syllabi, **p95 <15s**
- Designed **Supabase/Postgres** layer for **3K+** users with **RLS, JWT & RBAC** multi-tenant logic
- Deployed **Stripe** billing with trials & promos via an idempotency ledger eliminating webhook race conditions
- Built timezone-aware scheduler using **Cloud Run** to reliably process **13K+** batch email/push notifications

UW Orbital (Satellite Design Team)

Sep. 2025 – Apr. 2026

Backend Software Engineer, Ground Station

Waterloo, ON

- Built satellite comms recovery system (**FastAPI WebSockets**), regaining control of dropped sessions at **p95 <2s**
- Built single-threaded writer with operator leases, letting humans preempt a running script with **0** racing sends
- Persisted commands to a **write-ahead log** before each RF send, so mid-send crashes surface ambiguous outcomes
- Resolved every unacked command over **10K** sends by diffing ground log vs. **telemetry**, never double-firing
- Fenced zombie sockets with per-connection **epochs** validated server-side so reconnects never race its dead session

PROJECTS

RaceTime 🌀 | FastAPI, React, TypeScript, Redis, GitHub Actions, Kubernetes, Traefik, Prometheus, Grafana

- Built a real-time F1 race dashboard delivering live car positions via **FastAPI, React, Redis**, and **MQTT**
- Applied **cubic spline interpolation** to produce physics-like smooth car motion from discrete snapshots
- Prevented stutter by pacing playback at **220–280ms** based on Redis queue depth to absorb network jitter
- Deployed 7 services on **Kubernetes** with **Traefik HTTPS** ingress + **Prometheus/Grafana** observability